

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT
on

OPERATING SYSTEMS

Submitted by
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in partial fulfillment for the award of the degree of
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in
COMPUTER SCIENCE AND ENGINEERING



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CERTIFICATE

This is to certify that the Lab work entitled “OPERATING SYSTEMS – 23CS4PCOPS” carried out by ABHISHEK K S (1BM24CS010), who is Bonafide student of B. M. S. College of Engineering. It is in partial fulfilment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year Feb-2026 to June-2026.
The Lab report has been approved as it satisfies the academic requirements in respect of a OPERATING SYSTEMS - (23CS4PCOPS) work prescribed for the said degree.

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Course Outcomes

C01	Apply the different concepts and functionalities of Operating System
C02	Analyse various Operating system strategies and techniques
C03	Demonstrate the different functionalities of Operating System.
C04	Conduct practical experiments to implement the functionalities of Operating system.

Program 1a

Question: Write a C program to simulate the following non-pre-emptive CPU scheduling algorithm to find turnaround time and waiting time.

- FCFS
- SJF (pre-emptive & Non-preemptive)

Soln: =>FCFS

```
#include <stdio.h>
#include <stdlib.h>

struct process {
    int id, at, bt, ct, tat, wt, rt;
};

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct process p[n];
    for (int i = 0; i < n; i++) {
        p[i].id = i + 1;
        printf("Enter Arrival Time for P%d: ", p[i].id);
        scanf("%d", &p[i].at);
        printf("Enter Burst Time for P%d: ", p[i].id);
        scanf("%d", &p[i].bt);
    }
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (p[j].at > p[j + 1].at) {
                struct process temp = p[j];
                p[j] = p[j + 1];
                p[j + 1] = temp;
            }
        }
    }
    int timepassed = 0, sumtat = 0, sumwt = 0, sumrt = 0;
    for (int i = 0; i < n; i++) {
        if (timepassed < p[i].at) {
            timepassed = p[i].at;
        }
        p[i].ct = timepassed + p[i].bt;
        p[i].tat = p[i].ct - p[i].at;
        p[i].wt = p[i].tat - p[i].bt;

        timepassed += p[i].bt;

        sumtat += p[i].tat;
        sumwt += p[i].wt;
    }
    printf("\nID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
```

```

for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].id, p[i].at, p[i].bt, p[i].ct,
        p[i].tat, p[i].wt);
}
printf("\nAverage TAT: %.2f\n", (float)sumtat / n);
printf("Average WT: %.2f\n", (float)sumwt / n);

return 0;
}

```

```

Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\BMSCECSE-14>C:\Users\BMSCECSE-14\Desktop\18M2HCS010 Lab1 OS scheduling programs\CPU scheduling OS lab program 1.o"
C:\Users\BMSCECSE-14>C:\Users\BMSCECSE-14\Desktop\18M2HCS010 Lab1 OS scheduling programs\CPU scheduling OS lab program 1.exe"
Enter the number of processes: 4
Enter Arrival Time for P1: 0
Enter Burst Time for P1: 2
Enter Arrival Time for P2: 1
Enter Burst Time for P2: 2
Enter Arrival Time for P3: 5
Enter Burst Time for P3: 3
Enter Arrival Time for P4: 6
Enter Burst Time for P4: 4

ID   AT   BT   CT   TAT  WT   RT
1    0    2    2    2    0    0
2    1    2    4    3    1    1
3    5    3    8    3    0    0
4    6    4   12    6    2    2

Average TAT: 3.50
Average WT: 0.75

C:\Users\BMSCECSE-14>
Enter command number:

```

SOLN: =>SJF(Preemptive)

```
#include<stdio.h>
```

```
struct process {  
    int id, at, bt, ct, tat, wt, rt, rem;  
};
```

```
int main() {  
    int n;  
    printf("Enter the number of processes: ");  
    scanf("%d", &n);
```

```
    struct process p[n];  
    for (int i = 0; i < n; i++) {  
        p[i].id = i + 1;  
        printf("Enter Arrival Time for P%d: ", p[i].id);  
        scanf("%d", &p[i].at);  
        printf("Enter Burst Time for P%d: ", p[i].id);  
        scanf("%d", &p[i].bt);  
        p[i].rem = p[i].bt;  
        p[i].rt = -1;  
    }
```

```
    int completed = 0, time = 0, sumtat = 0, sumwt = 0, sumrt = 0;  
    while (completed != n) {  
        int idx = -1, minrem = 1e9;  
        for (int i = 0; i < n; i++) {  
            if (p[i].at <= time && p[i].rem > 0) {  
                if (p[i].rem < minrem || (p[i].rem == minrem && p[i].at < p[idx].at)) {  
                    minrem = p[i].rem;  
                    idx = i;  
                }  
            }  
        }  
        if (idx == -1) {  
            time++;  
            continue;  
        }  
        if (p[idx].rt == -1) {  
            p[idx].rt = time - p[idx].at;  
            sumrt += p[idx].rt;  
        }  
        p[idx].rem--;  
        time++;  
        if (p[idx].rem == 0) {  
            completed++;  
            p[idx].ct = time;  
            p[idx].tat = p[idx].ct - p[idx].at;  
            p[idx].wt = p[idx].tat - p[idx].bt;  
            sumtat += p[idx].tat;  
            sumwt += p[idx].wt;  
        }  
    }
```

```

printf("\nID\tAT\tBT\tCT\tTAT\tWT\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].id, p[i].at, p[i].bt, p[i].ct,
        p[i].tat, p[i].wt);
}

printf("\nAverage TAT: %.2f\n", (float)sumtat / n);
printf("Average WT: %.2f\n", (float)sumwt / n);
return 0;
}

```

```

Microsoft Windows [Version 10.0.22631.6199]
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C:\Users\BMSCECSE-14>C:\Users\BMSCECSE-14\Desktop\1BM24CS010 Abhishek K S Lab1 OS scheduling programs\CPU scheduling SJF(Preemptive).exe"
Enter the number of processes: 5
Enter Arrival Time for P1: 2
Enter Burst Time for P1: 1
Enter Arrival Time for P2: 1
Enter Burst Time for P2: 5
Enter Arrival Time for P3: 4
Enter Burst Time for P3: 1
Enter Arrival Time for P4: 0
Enter Burst Time for P4: 6
Enter Arrival Time for P5: 2
Enter Burst Time for P5: 3

ID    AT    BT    CT    TAT    WT
1     2     1     3     1     0
2     1     5    16    15    10
3     4     1     5     1     0
4     0     6    11    11     5
5     2     3     7     5     2

Average TAT: 6.60
Average WT: 3.40

C:\Users\BMSCECSE-14>

```

SOLN: => SJF Non preemptive

```
#include<stdio.h>
struct process{
    int id,at,bt,ct,tat,wt,rt;
};

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct process p[n];
    for (int i = 0; i < n; i++){
        p[i].id = i + 1;
        printf("Enter Arrival Time for P%d: ", p[i].id);
        scanf("%d", &p[i].at);
        printf("Enter Burst Time for P%d: ", p[i].id);
        scanf("%d", &p[i].bt);
    }

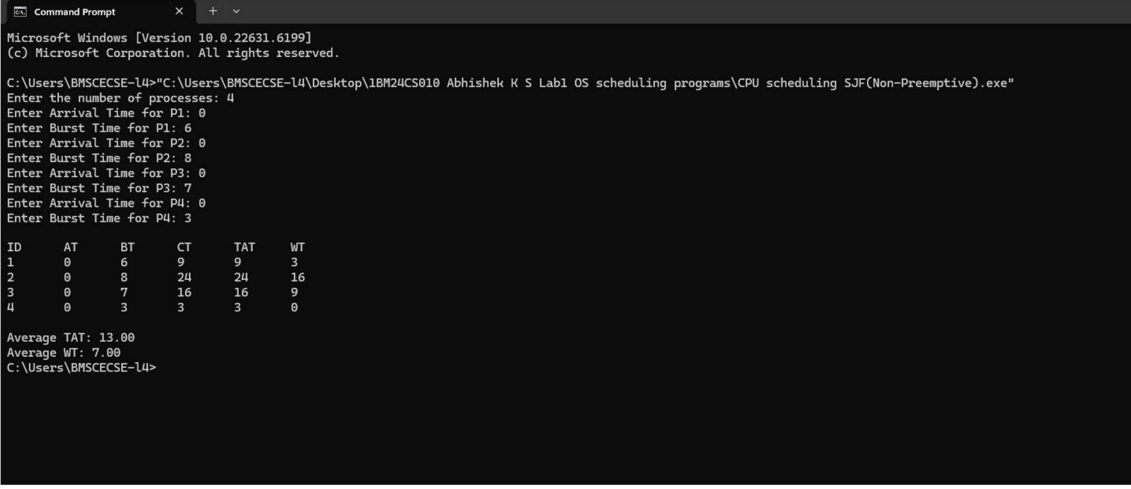
    int completed=0,time=0,sumtat=0,sumwt=0;
    int iscompleted[n];
    for(int i=0;i<n;i++) iscompleted[i]=0;

    while(completed!=n){
        int index=-1;
        int minbt=1e9;
        for(int i=0;i<n;i++){
            if(p[i].at<=time && !iscompleted[i]){
                if(p[i].bt<minbt){
                    minbt=p[i].bt;
                    index=i;
                }
            }
        }
        if(index!=-1){
            p[index].ct=time+p[index].bt;
            p[index].tat=p[index].ct-p[index].at;
            p[index].wt=p[index].tat-p[index].bt;
            p[index].rt=p[index].wt;
            sumtat+=p[index].tat;
            sumwt+=p[index].wt;
            time=p[index].ct;
            iscompleted[index]=1;
            completed++;
        } else {
            time++;
        }
    }

    printf("\nID\tAT\tBT\tCT\tTAT\tWT\n");
    for(int i=0;i<n;i++){
        printf("%d\t%d\t%d\t%d\t%d\t%d\n",p[i].id,p[i].at,p[i].bt,p[i].ct,p[i].tat,p[i].wt);
    }
}
```



```
printf("\nAverage TAT: %.2f", (float)sumtat/n);  
printf("\nAverage WT: %.2f", (float)sumwt/n);  
}
```



```
Microsoft Windows [Version 10.0.22631.6199]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\BMSCECSE-L4>"C:\Users\BMSCECSE-L4\Desktop\IBM24CS010 Abhishek K S Lab1 OS scheduling programs\CPU scheduling SJF(Non-Preemptive).exe"  
Enter the number of processes: 4  
Enter Arrival Time for P1: 0  
Enter Burst Time for P1: 6  
Enter Arrival Time for P2: 0  
Enter Burst Time for P2: 8  
Enter Arrival Time for P3: 0  
Enter Burst Time for P3: 7  
Enter Arrival Time for P4: 0  
Enter Burst Time for P4: 3  
  
ID    AT    BT    CT    TAT    WT  
1     0     6     9     9     3  
2     0     8    24    24    16  
3     0     7    16    16     9  
4     0     3     3     3     0  
  
Average TAT: 13.00  
Average WT: 7.00  
C:\Users\BMSCECSE-L4>
```

PROGRAM 1b

Question : Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time.

→ Priority (pre-emptive & Non-pre-emptive)

→ Round Robin (Experiment with different quantum sizes for RR algorithm)

SOLN: Priority Pre-emptive

```
#include <stdio.h>

struct process {
    int pid, at, bt, p, rt, ct, wt, tat;
};

int main() {
    int n, time = 0, completed = 0;
    printf("Enter the number of processes: ");
    scanf("%d", &n);
    struct process p[n];

    for (int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        printf("Enter the arrival time, burst time, priority: ");
        scanf("%d%d%d", &p[i].at, &p[i].bt, &p[i].p);
        p[i].rt = p[i].bt;
    }

    while (completed < n) {
        int idx = -1, highest = 9999;
        for (int i = 0; i < n; i++) {
            if (p[i].at <= time && p[i].rt > 0) {
                if (p[i].p < highest) {
                    highest = p[i].p;
                    idx = i;
                }
            }
        }
        if (idx != -1) {
            p[idx].rt--;
            time++;
            if (p[idx].rt == 0) {
                completed++;
                p[idx].ct = time;
                p[idx].tat = p[idx].ct - p[idx].at;
                p[idx].wt = p[idx].tat - p[idx].bt;
            }
        } else {
            time++;
        }
    }

    float totalwt = 0, totaltat = 0;
    printf("\nPid\tAt\tBt\tP\tCt\tTat\tWt\n");
```

```

for (int i = 0; i < n; i++) {
    totalwt += p[i].wt;
    totaltat += p[i].tat;
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt, p[i].p,
        p[i].ct, p[i].tat, p[i].wt);
}

printf("\nAverage waiting time : %.2f", totalwt / n);
printf("\nAverage turnaround time : %.2f\n", totaltat / n);

return 0;
}

```

```

PS C:\Users\BMSCECSE-14\Desktop\1BM24CS010 OS lab2 scheduling> & "C:\Users\BMSCECSE-14\Desktop\1BM24CS010 OS lab2 scheduling\Priority scheduling Preemptive.exe"
Enter the number of processes: 5
Enter the arrival time, burst time, priority: 0
3
3
Enter the arrival time, burst time, priority: 1
4
2
Enter the arrival time, burst time, priority: 2
6
4
Enter the arrival time, burst time, priority: 3
4
6
Enter the arrival time, burst time, priority: 5
2
10

```

Pid	At	Bt	P	Ct	Tat	Wt
1	0	3	3	7	7	4
2	1	4	2	5	4	0
3	2	6	4	13	11	5
4	3	4	6	17	14	10
5	5	2	10	19	14	12

```

Average waiting time : 6.20
Average turnaround time : 10.00
PS C:\Users\BMSCECSE-14\Desktop\1BM24CS010 OS lab2 scheduling> |

```

SOLN : Non pre-emptive

```
#include <stdio.h>

struct process {
    int id, at, bt, pr, ct, tat, wt, d;
};

int main() {
    int n, t = 0, c = 0, h;
    float tw = 0, tt = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct process p[n];

    for (int i = 0; i < n; i++) {
        p[i].id = i + 1;
        printf("Process %d\n", p[i].id);
        printf("Arrival time: ");
        scanf("%d", &p[i].at);
        printf("Burst time: ");
        scanf("%d", &p[i].bt);
        printf("Priority: ");
        scanf("%d", &p[i].pr);
        p[i].d = 0;
    }

    while (c != n) {
        h = -1;
        for (int i = 0; i < n; i++) {
            if (p[i].at <= t && p[i].d == 0) {
                if (h == -1 || p[i].pr < p[h].pr) {
                    h = i;
                }
            }
        }

        if (h == -1) {
            t++;
        } else {
            t += p[h].bt;
            p[h].ct = t;
            p[h].tat = p[h].ct - p[h].at;
            p[h].wt = p[h].tat - p[h].bt;
            tw += p[h].wt;
            tt += p[h].tat;
            p[h].d = 1;
            c++;
        }
    }

    printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");
    for (int i = 0; i < n; i++) {
```

```

        printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
            p[i].id, p[i].at, p[i].bt, p[i].pr,
            p[i].ct, p[i].tat, p[i].wt);
    }

    printf("\nAverage Turnaround Time: %.2f", tt / n);
    printf("\nAverage Waiting Time: %.2f\n", tw / n);

    return 0;
}

```

```

PS C:\Users\BMSCECSE-l4\Desktop\1BM24CS010 OS lab2 scheduling> & "C:\Users\BMSCECSE-l4\Desktop\1BM24CS010 OS lab2 scheduling\Priority scheduling Non-Preemptive.exe"
Enter number of processes: 5
Process 1
Arrival time: 0
Burst time: 3
Priority: 5
Process 2
Arrival time: 2
Burst time: 2
Priority: 3
Process 3
Arrival time: 3
Burst time: 5
Priority: 2
Process 4
Arrival time: 4
Burst time: 4
Priority: 4
Process 5
Arrival time: 6
Burst time: 1
Priority: 1

```

Process	AT	BT	PR	CT	TAT	WT
1	0	3	5	3	3	0
2	2	2	3	11	9	7
3	3	5	2	8	5	0
4	4	4	4	15	11	7
5	6	1	1	9	3	2

```

Average Turnaround Time: 6.20
Average Waiting Time: 3.20

```

SOLN : Round Robin

```
#include <stdio.h>

struct Process {
    int pid, at, bt, ct, tat, wt, rt;
};

int main() {
    int n, tq;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter Time Quantum: ");
    scanf("%d", &tq);

    struct Process p[n];
    int rem_bt[n];
    int readyQ[100];
    int front = 0, rear = 0;

    printf("Enter arrival times:\n");
    for(int i=0; i<n; i++) {
        scanf("%d", &p[i].at);
        p[i].pid = i+1;
    }

    printf("Enter burst times:\n");
    for(int i=0; i<n; i++) {
        scanf("%d", &p[i].bt);
        rem_bt[i] = p[i].bt;
        p[i].rt = p[i].bt;
        p[i].ct = p[i].tat = p[i].wt = 0;
    }

    int time = 0, completed = 0;
    int visited[n];
    for(int i=0; i<n; i++) visited[i] = 0;

    for(int i=0; i<n; i++) {
        if(p[i].at == 0) {
            readyQ[rear++] = i;
            visited[i] = 1;
        }
    }

    while(completed < n) {
        if(front == rear) {
            time++;
            for(int i=0; i<n; i++) {
                if(!visited[i] && p[i].at <= time) {
                    readyQ[rear++] = i;
                    visited[i] = 1;
                }
            }
        }
    }
}
```

```

    }
    continue;
}

int idx = readyQ[front++];
if(rem_bt[idx] > tq) {
    time += tq;
    rem_bt[idx] -= tq;
    p[idx].rt = rem_bt[idx];
} else {
    time += rem_bt[idx];
    rem_bt[idx] = 0;
    p[idx].ct = time;
    p[idx].tat = p[idx].ct - p[idx].at;
    p[idx].wt = p[idx].tat - p[idx].bt;
    p[idx].rt = 0;
    completed++;
}

for(int i=0; i<n; i++) {
    if(!visited[i] && p[i].at <= time) {
        readyQ[rear++] = i;
        visited[i] = 1;
    }
}

if(rem_bt[idx] > 0) {
    readyQ[rear++] = idx;
}
}

printf("\nRRS scheduling:\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
for(int i=0; i<n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt, p[i].ct,
        p[i].tat, p[i].wt);
}

float avg_TAT = 0, avg_WT = 0;
for(int i=0; i<n; i++) {
    avg_TAT += p[i].tat;
    avg_WT += p[i].wt;
}

printf("\nAverage turnaround time: %.2f ms\n", avg_TAT/n);
printf("Average waiting time: %.2f ms\n", avg_WT/n);

return 0;
}

```

```

PS C:\Users\BMSCECSE-14\Desktop\18M24CS010 OS lab2 scheduling> & "C:\Users\BMSCECSE-14\Desktop\18M24CS010 OS lab2 scheduling\Round Robin Scheduling.exe"
Enter number of processes: 5
Enter Time Quantum: 2
Enter arrival times:
0
1
2
3
4
Enter burst times:
5
3
1
2
3

RRS scheduling:


| PID | AT | BT | CT | TAT | WT |
|-----|----|----|----|-----|----|
| 1   | 0  | 5  | 13 | 13  | 8  |
| 2   | 1  | 3  | 12 | 11  | 8  |
| 3   | 2  | 1  | 5  | 3   | 2  |
| 4   | 3  | 2  | 9  | 6   | 4  |
| 5   | 4  | 3  | 14 | 10  | 7  |



Average turnaround time: 8.60 ms
Average waiting time: 5.60 ms
PS C:\Users\BMSCECSE-14\Desktop\18M24CS010 OS lab2 scheduling> |

```


PROGRAM 2

Question : Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.

SOLN :

```
#include <stdio.h>

#define MAX 10

typedef struct {
    int pid, at, bt, rt, ct, tat, wt, queue;
} Process;

int main() {
    Process p[MAX];
    int n, i, time = 0, completed = 0;
    int tq = 2;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("\nP%d\n", i + 1);
        p[i].pid = i + 1;

        printf("Arrival Time: ");
        scanf("%d", &p[i].at);

        printf("Burst Time: ");
        scanf("%d", &p[i].bt);

        printf("Queue (1-System, 2-User): ");
        scanf("%d", &p[i].queue);

        p[i].rt = p[i].bt;
    }

    printf("\nGantt Chart:\n");

    while (completed < n) {

        int executed = 0;

        for (i = 0; i < n; i++) {
            if (p[i].queue == 1 && p[i].rt > 0 && p[i].at <= time) {

                executed = 1;

                if (p[i].rt > tq) {
                    printf("P%d ", p[i].pid);
                    time += tq;
                }
            }
        }
    }
}
```

```

p[i].rt -= tq;
} else {
printf("P%d ", p[i].pid);
time += p[i].rt;
p[i].rt = 0;
p[i].ct = time;
completed++;
}
}
}

if (!executed) {
for (i = 0; i < n; i++) {
if (p[i].queue == 2 && p[i].rt > 0 && p[i].at <= time) {

executed = 1;

printf("P%d ", p[i].pid);
p[i].rt--;
time++;

if (p[i].rt == 0) {
p[i].ct = time;
completed++;
}

break;
}
}
}

if (!executed) {
time++;
}
}

float total_tat = 0, total_wt = 0;

printf("\n\nPID\tAT\tBT\tCT\tTAT\tWT\n");

for (i = 0; i < n; i++) {
p[i].tat = p[i].ct - p[i].at;
p[i].wt = p[i].tat - p[i].bt;

total_tat += p[i].tat;
total_wt += p[i].wt;

printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt,
        p[i].ct, p[i].tat, p[i].wt);
}

printf("\nAverage TAT = %.2f", total_tat / n);
printf("\nAverage WT = %.2f\n", total_wt / n);

```

```

return 0;
}

```

```

PS C:\Users\BMSCECSE-l4\Desktop\1BM24CS010 OS lab program1> & "C:\Users\BMSCECSE-l4\Desktop\1BM24CS010 OS lab program1\Multilevel Queue Program.exe"
Enter number of processes: 4

P1
Arrival Time: 0
Burst Time: 4
Queue (1-System, 2-User): 1

P2
Arrival Time: 0
Burst Time: 3
Queue (1-System, 2-User): 1

P3
Arrival Time: 0
Burst Time: 8
Queue (1-System, 2-User): 2

P4
Arrival Time: 10
Burst Time: 5
Queue (1-System, 2-User): 1

Gantt Chart:
P1 P2 P1 P2 P3 P3 P4 P4 P4 P3 P3 P3 P3

PID   AT   BT   CT   TAT  WT
P1     0    4    6    6    2
P2     0    3    7    7    4
P3     0    8   20   20   12
P4    10    5   15    5    0

Average TAT = 9.50
Average WT = 4.50
PS C:\Users\BMSCECSE-l4\Desktop\1BM24CS010 OS lab program1> |

```

PROGRAM 3

QUESTION : Write a C program to simulate Real-Time CPU Scheduling algorithms:

Rate- Monotonic

Earliest-deadline First

Proportional scheduling

SOLN : Rate Monotonic Scheduling

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
struct Process {
    int pid;
    int burst;
    int period;
    int remaining;
};

int gcd(int a, int b) {
    if (b == 0) return a;
    return gcd(b, a % b);
}
int lcm(int a, int b) {
    return (a * b) / gcd(a, b);
}

int main() {
    int n;
    printf("Enter the number of processes: ");
    scanf("%d", &n);

    struct Process p[n];
    printf("Enter the CPU burst times:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &p[i].burst);
        p[i].pid = i + 1;
    }

    printf("Enter the time periods:\n");
    int hyperPeriod = 1;
    for (int i = 0; i < n; i++) {
        scanf("%d", &p[i].period);
        hyperPeriod = lcm(hyperPeriod, p[i].period);
    }

    printf("\nLCM=%d\n", hyperPeriod);
```

```

printf("\nRate Monotone Scheduling:\n");
printf("PID\tBurst\tPeriod\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\n", p[i].pid, p[i].burst, p[i].period);
}

double U = 0.0;
for (int i = 0; i < n; i++) {
    U += (double)p[i].burst / p[i].period;
}
double bound = n * (pow(2.0, 1.0/n) - 1);
printf("\n%.6f <= %.6f => %s\n", U, bound, (U <= bound ? "true" : "false"));
printf("Scheduling occurs for %d ms\n", hyperPeriod);

for (int t = 0; t < hyperPeriod; t++) {
    int chosen = -1;

    for (int i = 0; i < n; i++) {
        if (t % p[i].period == 0) {
            p[i].remaining = p[i].burst;
        }
    }

    for (int i = 0; i < n; i++) {
        if (p[i].remaining > 0) {
            chosen = i;
            break;
        }
    }

    if (chosen != -1) {
        printf("%dms onwards: Process %d running\n", t, p[chosen].pid);
        p[chosen].remaining--;
    } else {
        printf("%dms onwards: CPU is idle\n", t);
    }
}

return 0;
}

```

```
PS C:\Users\BMSCECSE-l4\Desktop\1BM24CS010 Abhishek K S OS lab program 3> & "C:\Users\BMSCECSE-l4\Desktop\1BM24CS010 Abhishek K S OS lab program 3\Rate Monotonic Scheduling.exe"
Enter the number of processes: 2
Enter the CPU burst times:
20
35
Enter the time periods:
50
100

LCM=100

Rate Monotone Scheduling:
PID    Burst  Period
1      20    50
2      35    100

0.750000 <= 0.828427 ==> true
Scheduling occurs for 100 ms

0ms onwards: Process 1 running
1ms onwards: Process 1 running
2ms onwards: Process 1 running
3ms onwards: Process 1 running
4ms onwards: Process 1 running
```

SOLN : Earliest Deadline First

```
#include <stdio.h>
```

```
#include <limits.h>
```

```
#define MAX_TASKS 10
```

```
typedef struct {
```

```
    int pid;
```

```
    int burst;
```

```
    int deadline;
```

```
    int period;
```

```
} Task;
```

```
int gcd(int a, int b) {
```

```
    while (b != 0) {
```

```
        int temp = b;
```

```
        b = a % b;
```

```
        a = temp;
```

```
    }
```

```
    return a;
```

```
}
```

```
int lcm(int a, int b) {
```

```
    return (a * b) / gcd(a, b);
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    Task tasks[MAX_TASKS];
```

```
    printf("Enter number of tasks: ");
```

```
    scanf("%d", &n);
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("\nTask %d:\n", i + 1);
```

```
        tasks[i].pid = i + 1;
```

```
        printf("Enter Burst Time: ");
```

```
        scanf("%d", &tasks[i].burst);
```

```
        printf("Enter Deadline: ");
```

```
        scanf("%d", &tasks[i].deadline);
```

```
        printf("Enter Period: ");
```

```
        scanf("%d", &tasks[i].period);
```

```
    }
```

```
    int SIMULATION_TIME = tasks[0].period;
```

```

for (int i = 1; i < n; i++) {
    SIMULATION_TIME = lcm(SIMULATION_TIME, tasks[i].period);
}

int remaining[MAX_TASKS] = {0};
int next_release[MAX_TASKS] = {0};
int abs_deadline[MAX_TASKS] = {0};
int active[MAX_TASKS] = {0};

printf("\nScheduling occurs for %d ms (LCM of periods)\n\n", SIMULATION_TIME);

for (int t = 0; t < SIMULATION_TIME; t++) {

    // Release jobs
    for (int i = 0; i < n; i++) {
        if (t == next_release[i]) {
            remaining[i] = tasks[i].burst;
            abs_deadline[i] = t + tasks[i].deadline;
            next_release[i] += tasks[i].period;
            active[i] = 1;
        }
    }

    int chosen = -1;
    int earliest = INT_MAX;

    for (int i = 0; i < n; i++) {
        if (active[i] && remaining[i] > 0) {
            if (abs_deadline[i] < earliest) {
                earliest = abs_deadline[i];
                chosen = i;
            }
        }
    }

    if (chosen != -1) {
        printf("%dms : Task %d is running.\n", t, tasks[chosen].pid);
        remaining[chosen]--;

        if (remaining[chosen] == 0) {
            active[chosen] = 0;
        }
    } else {
        printf("%dms : CPU is idle.\n", t);
    }
}

```



```
    return 0;
}
```

```
PS C:\Users\BMSCECSE-14\Desktop\IBM24CS010 Abhishek K S OS Lab program 3> & "C:\Users\BMSCECSE-14\Desktop\IBM24CS010 Abhishek K S OS Lab program 3\Earliest Deadline First.exe"
Enter number of tasks: 3

Task 1:
Enter Burst Time: 2
Enter Deadline: 4
Enter Period: 5

Task 2:
Enter Burst Time: 3
Enter Deadline: 7
Enter Period: 20

Task 3:
Enter Burst Time: 2
Enter Deadline: 8
Enter Period: 10

Scheduling occurs for 20 ms (LCM of periods)

0ms : Task 1 is running.
1ms : Task 1 is running.
2ms : Task 2 is running.
3ms : Task 2 is running.
4ms : Task 2 is running.
5ms : Task 3 is running.
6ms : Task 3 is running.
```

PROGRAM 4 :

QUESTION :

- 4a. Write a C program to simulate producer-consumer problem using semaphores
- 4b. Write a C program to simulate the concept of Dining Philosophers problem

SOLN : Bounded Buffer

```
#include <stdio.h>
#include <stdlib.h>
#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;

int empty = BUFFER_SIZE;
int full = 0;

void produce() {
    if (empty == 0) {
        printf("\nBuffer is full! Cannot produce.\n");
        return;
    }

    int item;
    printf("Enter item to produce: ");
    scanf("%d", &item);

    buffer[in] = item;
    printf("Produced: %d\n", item, in);

    in = (in + 1) % BUFFER_SIZE;
    empty--;
    full++;
}

void consume() {
    if (full == 0) {
        printf("Buffer is empty! Cannot consume.\n");
        return;
    }

    int item = buffer[out];
    printf("Consumed: %d\n", item, out);

    out = (out + 1) % BUFFER_SIZE;
```

```

    full--;
    empty++;
}

int main() {
    int choice;
    printf("Buffer Size: %d\n", BUFFER_SIZE);

    while (1) {
        printf("\n1. Produce\n2. Consume\n3. Exit\n");
        printf("Current State: [Full slots: %d | Empty slots: %d]\n", full, empty);
        printf("Enter choice: ");

        if (scanf("%d", &choice) != 1) break;

        switch (choice) {
            case 1:
                produce();
                break;
            case 2:
                consume();
                break;
            case 3:
                printf("Exiting...\n");
                exit(0);
            default:
                printf("Invalid choice!\n");
        }
    }

    return 0;
}

```

```
IBM24CS010 ABHISHEK K S
Buffer Size: 5

1. Produce
2. Consume
3. Exit
Current State: [Full slots: 0 | Empty slots: 5]
Enter choice: 1
Enter item to produce: 10
Produced: 10

1. Produce
2. Consume
3. Exit
Current State: [Full slots: 1 | Empty slots: 4]
Enter choice: 1
Enter item to produce: 20
Produced: 20

1. Produce
2. Consume
3. Exit
Current State: [Full slots: 2 | Empty slots: 3]
Enter choice: 1
Enter item to produce: 30
Produced: 30

1. Produce
2. Consume
3. Exit
Current State: [Full slots: 3 | Empty slots: 2]
Enter choice: 2
Consumed: 10

1. Produce
2. Consume
3. Exit
Current State: [Full slots: 2 | Empty slots: 3]
Enter choice: 10
Invalid choice!

1. Produce
2. Consume
3. Exit
Current State: [Full slots: 2 | Empty slots: 3]
Enter choice: 2
Consumed: 20
```

SOLN : Dining Philosopher

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
#define N 5
pthread_mutex_t forks[N];
pthread_t philosophers[N];

void* philosopher(void* num)
{
    int id = *(int*)num;
    int left = id; // left fork index
    int right = (id + 1) % N; // right fork index
    while (1)
    {
        printf("Philosopher %d is thinking.\n", id);
        sleep(1); // Thinking
        // To avoid deadlock, philosophers with even id pick left then right,
        // odd id pick right then left.
        if (id % 2 == 0)
        {
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked up left fork %d.\n", id, left);
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked up right fork %d.\n", id, right); }
        else
        {
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked up right fork %d.\n", id, right); // Pick up left
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked up left fork %d.\n", id, left);
        }
        printf("Philosopher %d is eating.\n", id);
        sleep(2); // Put down forks
        pthread_mutex_unlock(&forks[left]);
        pthread_mutex_unlock(&forks[right]);
        printf("Philosopher %d put down forks %d and %d.\n", id, left, right);
    }
    return NULL;
}

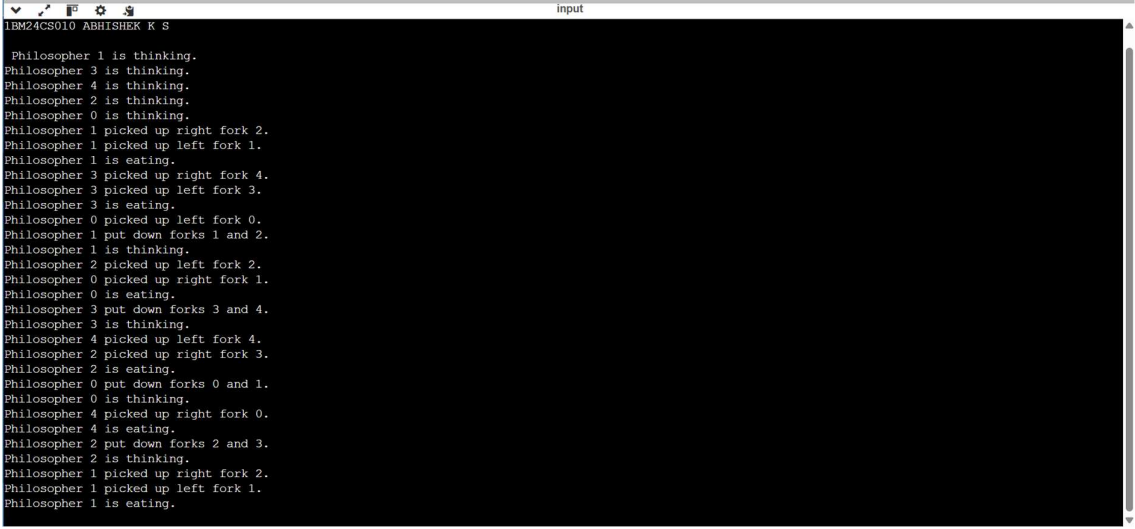
int main() {
    int i;
    int ids[N]; // Initialize forks (mutexes)
    for (i = 0; i < N; i++)
```

```

{ pthread_mutex_init(&forks[i], NULL);
ids[i] = i;
}
for (i = 0; i < N; i++){
pthread_create(&philosophers[i], NULL, philosopher, &ids[i]);
}

for (i = 0; i < N; i++){
pthread_join(philosophers[i], NULL); }
for (i = 0; i < N; i++){
pthread_mutex_destroy(&forks[i]);
}
return 0;
}
}

```



```

input
1RM24CS010 ABHISHEK K S

Philosopher 1 is thinking.
Philosopher 3 is thinking.
Philosopher 4 is thinking.
Philosopher 2 is thinking.
Philosopher 0 is thinking.
Philosopher 1 picked up right fork 2.
Philosopher 1 picked up left fork 1.
Philosopher 1 is eating.
Philosopher 3 picked up right fork 4.
Philosopher 3 picked up left fork 3.
Philosopher 3 is eating.
Philosopher 0 picked up left fork 0.
Philosopher 1 put down forks 1 and 2.
Philosopher 1 is thinking.
Philosopher 2 picked up left fork 2.
Philosopher 0 picked up right fork 1.
Philosopher 0 is eating.
Philosopher 3 put down forks 3 and 4.
Philosopher 3 is thinking.
Philosopher 4 picked up left fork 4.
Philosopher 2 picked up right fork 3.
Philosopher 2 is eating.
Philosopher 0 put down forks 0 and 1.
Philosopher 0 is thinking.
Philosopher 4 picked up right fork 0.
Philosopher 4 is eating.
Philosopher 2 put down forks 2 and 3.
Philosopher 2 is thinking.
Philosopher 1 picked up right fork 2.
Philosopher 1 picked up left fork 1.
Philosopher 1 is eating.

```

PROGRAM 5

QUESTION : Write a C program to simulate Bankers algorithm and deadlock avoidance

SOLN : Banker algorithm

```
#include <stdio.h>
#include <stdbool.h>

#define MAX_PROCESSES 10
#define MAX_RESOURCES 10

int main() {
    int n, m;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resource types: ");
    scanf("%d", &m);

    int allocation[MAX_PROCESSES][MAX_RESOURCES];
    int max[MAX_PROCESSES][MAX_RESOURCES];
    int need[MAX_PROCESSES][MAX_RESOURCES];
    int available[MAX_RESOURCES];

    printf("\nEnter Allocation Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("\nEnter Max Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &max[i][j]);
        }
    }

    printf("\nEnter Available Resources:\n");
    for (int j = 0; j < m; j++) {
        scanf("%d", &available[j]);
    }

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
```

```

        need[i][j] = max[i][j] - allocation[i][j];
    }
}

bool finish[MAX_PROCESSES] = {false};
int safeSequence[MAX_PROCESSES];

int work[MAX_RESOURCES];

// Work = Available
for (int j = 0; j < m; j++) {
    work[j] = available[j];
}

int count = 0;

while (count < n) {
    bool found = false;

    for (int i = 0; i < n; i++) {

        if (finish[i] == false) {

            bool canExecute = true;

            for (int j = 0; j < m; j++) {
                if (need[i][j] > work[j]) {
                    canExecute = false;
                    break;
                }
            }

            if (canExecute) {

                for (int j = 0; j < m; j++) {
                    work[j] += allocation[i][j];
                }

                safeSequence[count] = i;
                count++;

                finish[i] = true;
                found = true;
            }
        }
    }
}

```



```

        if (!found) {
            break;
        }
    }

    if (count == n) {
        printf("\nSystem is in SAFE state.\n");
        printf("Safe Sequence: ");

        for (int i = 0; i < n; i++) {
            printf("P%d", safeSequence[i]);

            if (i != n - 1) {
                printf(" -> ");
            }
        }

        printf("\n");
    }
    else {
        printf("\nSystem is NOT in safe state.\n");
    }

    return 0;
}

```

```

IBM24CS010 ABHISHEK K S

Enter number of processes: 5
Enter number of resource types: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

Enter Max Matrix:
7 5 3
3
2
2
9 0 2
2 2 2
4 3 3

Enter Available Resources:
3 3 2

System is in SAFE state.
Safe Sequence: P1 -> P3 -> P4 -> P0 -> P2

```

SOLN : Deadlock avoidance algorithm

```
#include <stdio.h>
#include <stdbool.h>

#define MAX_PROCESSES 10
#define MAX_RESOURCES 10

int main() {
    int n, m;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resource types: ");
    scanf("%d", &m);

    int allocation[MAX_PROCESSES][MAX_RESOURCES];
    int request[MAX_PROCESSES][MAX_RESOURCES];
    int available[MAX_RESOURCES];

    printf("\nEnter Allocation Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("\nEnter Request Matrix:\n");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            scanf("%d", &request[i][j]);
        }
    }

    printf("\nEnter Available Resources:\n");
    for (int j = 0; j < m; j++) {
        scanf("%d", &available[j]);
    }

    bool finish[MAX_PROCESSES];
    int work[MAX_RESOURCES];
    int safeSequence[MAX_PROCESSES];

    for (int j = 0; j < m; j++) {
        work[j] = available[j];
    }
```

```

for (int i = 0; i < n; i++) {
    bool zeroAllocation = true;

    for (int j = 0; j < m; j++) {
        if (allocation[i][j] != 0) {
            zeroAllocation = false;
            break;
        }
    }

    finish[i] = zeroAllocation;
}

int count = 0;

while (1) {
    bool found = false;

    for (int i = 0; i < n; i++) {

        if (finish[i] == false) {

            bool canExecute = true;

            for (int j = 0; j < m; j++) {
                if (request[i][j] > work[j]) {
                    canExecute = false;
                    break;
                }
            }

            if (canExecute) {

                for (int j = 0; j < m; j++) {
                    work[j] += allocation[i][j];
                }

                finish[i] = true;
                safeSequence[count] = i;
                count++;

                found = true;
            }
        }
    }
}

```

```

        if (!found) {
            break;
        }
    }

    bool deadlock = false;

    for (int i = 0; i < n; i++) {
        if (finish[i] == false) {
            deadlock = true;
            break;
        }
    }

    if (!deadlock) {
        printf("\nNo Deadlock detected.\n");

        printf("Safe Sequence: ");
        for (int i = 0; i < count; i++) {
            printf("P%d", safeSequence[i]);

            if (i != count - 1) {
                printf(" -> ");
            }
        }

        printf("\n");
    }
    else {
        printf("\nDeadlock detected.\n");

        printf("Processes involved in deadlock: ");

        for (int i = 0; i < n; i++) {
            if (finish[i] == false) {
                printf("P%d ", i);
            }
        }

        printf("\n");
    }

    return 0;
}

```

```
IBM24CS010 ABHISHEK K S

Enter number of processes: 5
Enter number of resource types: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 3
2 1 1
0 0 2

Enter Request Matrix:
0 0 0
2 0 2
0 0 0
1 0 0
0 0 2

Enter Available Resources:
000
0 0

No Deadlock detected.
Safe Sequence: P0 -> P2 -> P3 -> P4 -> P1
```

PROGRAM 6 :

QUESTION : Write a C program to simulate Memory Allocation (First Fit, Best Fit, Worst Fit)

SOLN : Memory Allocation

```
#include <stdio.h>
```

```
void firstFit(int blockSize[], int m, int processSize[], int n) {  
    int allocation[n];
```

```
    for (int i = 0; i < n; i++)  
        allocation[i] = -1;
```

```
    for (int i = 0; i < n; i++) {  
        for (int j = 0; j < m; j++) {  
            if (blockSize[j] >= processSize[i]) {  
                allocation[i] = j;  
                blockSize[j] = 0;  
                break;  
            }  
        }  
    }  
}
```

```
printf("\nFirst Fit Allocation:\n");  
printf("Process No.\tProcess Size\tBlock No.\n");
```

```
for (int i = 0; i < n; i++) {  
    printf("%d\t%d\t", i + 1, processSize[i]);
```

```
    if (allocation[i] != -1)  
        printf("%d\n", allocation[i] + 1);  
    else  
        printf("Not Allocated\n");  
}
```

```
}
```

```
void bestFit(int blockSize[], int m, int processSize[], int n) {  
    int allocation[n];
```

```
    for (int i = 0; i < n; i++)  
        allocation[i] = -1;
```

```
    for (int i = 0; i < n; i++) {  
        int bestIdx = -1;
```

```
        for (int j = 0; j < m; j++) {
```

```

        if (blockSize[j] >= processSize[i]) {
            if (bestIdx == -1 || blockSize[j] < blockSize[bestIdx]) {
                bestIdx = j;
            }
        }
    }

    if (bestIdx != -1) {
        allocation[i] = bestIdx;
        blockSize[bestIdx] = 0;
    }
}

printf("\nBest Fit Allocation:\n");
printf("Process No.\tProcess Size\tBlock No.\n");

for (int i = 0; i < n; i++) {
    printf("%d\t\t%d\t\t", i + 1, processSize[i]);

    if (allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}

}

void worstFit(int blockSize[], int m, int processSize[], int n) {
    int allocation[n];

    for (int i = 0; i < n; i++)
        allocation[i] = -1;

    for (int i = 0; i < n; i++) {
        int worstIdx = -1;

        for (int j = 0; j < m; j++) {
            if (blockSize[j] >= processSize[i]) {
                if (worstIdx == -1 || blockSize[j] > blockSize[worstIdx]) {
                    worstIdx = j;
                }
            }
        }

        if (worstIdx != -1) {
            allocation[i] = worstIdx;
            blockSize[worstIdx] = 0;
        }
    }
}

```

```

    }

    printf("\nWorst Fit Allocation:\n");
    printf("Process No.\tProcess Size\tBlock No.\n");

    for (int i = 0; i < n; i++) {
        printf("%d\t%d\t", i + 1, processSize[i]);

        if (allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

int main() {
    int m, n;

    printf("Enter number of memory blocks: ");
    scanf("%d", &m);

    int blocks[m], firstBlocks[m], bestBlocks[m], worstBlocks[m];

    printf("Enter sizes of memory blocks:\n");
    for (int i = 0; i < m; i++) {
        scanf("%d", &blocks[i]);
        firstBlocks[i] = blocks[i];
        bestBlocks[i] = blocks[i];
        worstBlocks[i] = blocks[i];
    }

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int processSize[n];

    printf("Enter sizes of processes:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &processSize[i]);
    }

    firstFit(firstBlocks, m, processSize, n);
    bestFit(bestBlocks, m, processSize, n);
    worstFit(worstBlocks, m, processSize, n);

    return 0;
}

```



```

PS C:\Users\BMSCECSE-SH\Desktop\Abhishek K S 18M24CS010 OS Lab Memory> & "C:\Users\BMSCECSE-SH\Desktop\Abhishek K S 18M24CS010 OS Lab Memory\MemoryAllocation.exe"
Enter number of memory blocks: 5
Enter sizes of memory blocks:
100
500
200
300
600
Enter number of processes: 4
Enter sizes of processes:
212
412
112
426

First Fit Allocation:
Process No.    Process Size    Block No.
1              212            2
2              412            5
3              112            3
4              426            Not Allocated

Best Fit Allocation:
Process No.    Process Size    Block No.
1              212            4
2              412            2
3              112            3
4              426            5

Worst Fit Allocation:
Process No.    Process Size    Block No.
1              212            5
2              412            2
3              112            4
4              426            Not Allocated
PS C:\Users\BMSCECSE-SH\Desktop\Abhishek K S 18M24CS010 OS Lab Memory>

```

PROGRAM 7 :

QUESTION : Write a C program to simulate Page Replacement

```
#include <stdio.h>
#include <stdbool.h>

#define MAX_PAGES 100

void printFrames(int frames[], int num_frames) {
    printf("[");
    int count = 0;
    for (int i = 0; i < num_frames; i++) {
        if (frames[i] != -1) {
            printf("%d", frames[i]);
            count++;
        }
    }
    printf("]\n");
}

void fifoPageReplacement(int pages[], int num_pages, int num_frames) {
    printf("\n--- FIFO Page Replacement ---\n");

    int frames[num_frames];
    for (int i = 0; i < num_frames; i++) frames[i] = -1;

    int page_faults = 0;
    int next = 0;

    for (int i = 0; i < num_pages; i++) {
        int page = pages[i];
        bool is_present = false;

        for (int j = 0; j < num_frames; j++) {
            if (frames[j] == page) {
                is_present = true;
                break;
            }
        }

        printf("Page %d -> ", page);

        if (!is_present) {
```

```

        frames[next] = page;
        next = (next + 1) % num_frames;
        page_faults++;
    }

    printFrames(frames, num_frames);
}

printf("Total Page Faults (FIFO): %d\n", page_faults);
}

void lruPageReplacement(int pages[], int num_pages, int num_frames) {
    printf("\n--- LRU Page Replacement ---\n");

    int frames[num_frames];
    int last_used[num_frames];
    for (int i = 0; i < num_frames; i++) {
        frames[i] = -1;
        last_used[i] = -1;
    }

    int page_faults = 0;

    for (int i = 0; i < num_pages; i++) {
        int page = pages[i];
        bool is_present = false;
        int present_index = -1;

        for (int j = 0; j < num_frames; j++) {
            if (frames[j] == page) {
                is_present = true;
                present_index = j;
                break;
            }
        }

        printf("Page %d -> ", page);

        if (is_present) {
            last_used[present_index] = i;
        } else {
            page_faults++;
        }
    }
}

```

```

    int empty_index = -1;
    for (int j = 0; j < num_frames; j++) {
        if (frames[j] == -1) {
            empty_index = j;
            break;
        }
    }

    if (empty_index != -1) {

        frames[empty_index] = page;
        last_used[empty_index] = i;
    } else {

        int lru_index = 0;
        int min_time = last_used[0];
        for (int j = 1; j < num_frames; j++) {
            if (last_used[j] < min_time) {
                min_time = last_used[j];
                lru_index = j;
            }
        }
        frames[lru_index] = page;
        last_used[lru_index] = i;
    }
}

printFrames(frames, num_frames);
}
printf("Total Page Faults (LRU): %d\n", page_faults);
}

void optimalPageReplacement(int pages[], int num_pages, int num_frames) {
    printf("\n--- Optimal Page Replacement ---\n");

    int frames[num_frames];
    for (int i = 0; i < num_frames; i++) frames[i] = -1;

    int page_faults = 0;

    for (int i = 0; i < num_pages; i++) {
        int page = pages[i];
        bool is_present = false;

        for (int j = 0; j < num_frames; j++) {
            if (frames[j] == page) {

```

```

        is_present = true;
        break;
    }
}

printf("Page %d -> ", page);

if (!is_present) {
    page_faults++;

    int empty_index = -1;
    for (int j = 0; j < num_frames; j++) {
        if (frames[j] == -1) {
            empty_index = j;
            break;
        }
    }

    if (empty_index != -1) {
        frames[empty_index] = page;
    } else {

        int replace_index = -1;
        int farthest_use = -1;

        for (int j = 0; j < num_frames; j++) {
            int next_use = -1;

            for (int k = i + 1; k < num_pages; k++) {
                if (pages[k] == frames[j]) {
                    next_use = k;
                    break;
                }
            }

            if (next_use == -1) {
                replace_index = j;
                break;
            }

            if (next_use > farthest_use) {
                farthest_use = next_use;
                replace_index = j;
            }
        }
    }
}

```

```

        frames[replace_index] = page;
    }
}

    printFrames(frames, num_frames);
}
printf("Total Page Faults (Optimal): %d\n", page_faults);
}

int main() {
    int num_pages, num_frames;
    int pages[MAX_PAGES];

    printf("Enter number of pages: ");
    if (scanf("%d", &num_pages) != 1 || num_pages <= 0 || num_pages > MAX_PAGES) {
        printf("Invalid number of pages.\n");
        return 1;
    }

    printf("Enter page reference string:\n");
    for (int i = 0; i < num_pages; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    if (scanf("%d", &num_frames) != 1 || num_frames <= 0) {
        printf("Invalid number of frames.\n");
        return 1;
    }

    fifoPageReplacement(pages, num_pages, num_frames);
    lruPageReplacement(pages, num_pages, num_frames);
    optimalPageReplacement(pages, num_pages, num_frames);

    return 0;
}

```

```

PS C:\Users\BMSCECSE-14\Desktop\OS Lab7 Page replacement Abhishek ks 18M24CS010> . "C:\Users\BMSCECSE-14\Desktop\OS Lab7 Page replacement Abhishek ks 18M24CS010\PageRep.exe"
Enter number of pages: 21
Enter page reference string:
5
0
1
0
2
3
0
2
4
0
3
2
0
0
1
2
7
0
1
1
0
Enter number of frames: 3

--- FIFO Page Replacement ---
Page 5 -> [5]
Page 0 -> [50]
Page 1 -> [501]
Page 0 -> [501]
Page 2 -> [201]
Page 3 -> [211]
Page 0 -> [210]
Page 2 -> [210]
Page 4 -> [410]
Page 1 -> [410]
Page 1 -> [410]
Page 2 -> [420]
Page 0 -> [420]
Page 2 -> [420]
Page 1 -> [421]
Page 2 -> [421]
Page 7 -> [721]
Page 0 -> [701]
Page 1 -> [701]
Page 1 -> [701]
Page 0 -> [701]
Total Page Faults (FIFO): 11

--- LRU Page Replacement ---
Page 5 -> [5]
Page 0 -> [50]

```

```

--- LRU Page Replacement ---
Page 5 -> [5]
Page 0 -> [50]
Page 1 -> [501]
Page 0 -> [501]
Page 2 -> [201]
Page 3 -> [203]
Page 0 -> [203]
Page 2 -> [203]
Page 4 -> [204]
Page 3 -> [230]
Page 3 -> [230]
Page 2 -> [230]
Page 0 -> [230]
Page 2 -> [230]
Page 1 -> [210]
Page 2 -> [210]
Page 7 -> [217]
Page 0 -> [207]
Page 1 -> [107]
Page 1 -> [107]
Page 0 -> [107]
Total Page Faults (LRU): 12

--- Optimal Page Replacement ---
Page 5 -> [5]
Page 0 -> [50]
Page 1 -> [501]
Page 0 -> [501]
Page 2 -> [201]
Page 3 -> [203]
Page 0 -> [203]
Page 2 -> [203]
Page 4 -> [203]
Page 3 -> [203]
Page 3 -> [203]
Page 2 -> [203]
Page 0 -> [203]
Page 2 -> [203]
Page 1 -> [201]
Page 2 -> [201]
Page 7 -> [701]
Page 0 -> [701]
Page 1 -> [701]
Page 1 -> [701]
Page 0 -> [701]
Total Page Faults (Optimal): 9
PS C:\Users\BMSCECSE-14\Desktop\OS Lab7 Page replacement Abhishek ks 18M24CS010>

```